

THE LEGAL ENGINEER'S FIELD GUIDE TO //

COMPUTA- TIONAL LEGAL METHODS



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DWG NO.	SCALE:	SHEET:
CLM-2026 - REV 00	NTS	1 OF 1
MATERIAL: CASE LAW + CURIOSITY		

Today's field guide

01 Introduction

what CLM is, and why use it

02 Quantitative text analysis

tf-idf, doc2vec, cosine

03 Network analysis

ego network, jaccard, communities

04 LLM methods

compare results, prompt an LLM

A problem stated in 1973

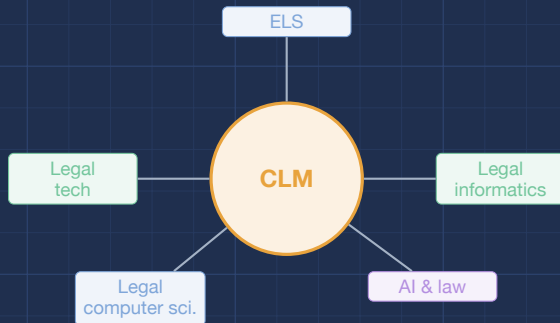
“Some doubt the ability of computers to solve the problem [of case retrieval] as a matter of theory... Others doubt the ability of computers to solve the problems as a matter of practice... Some doubt even the advisability of attempting to solve the problem.”

—Collin Tapper, Computers and the Law (1973), p. 106

- We require computers to make sense of vast and complex case law.
- Machine performance is approaching human level on many sub-tasks.
- The obstacles are now as much *legal-theoretical* as technical.

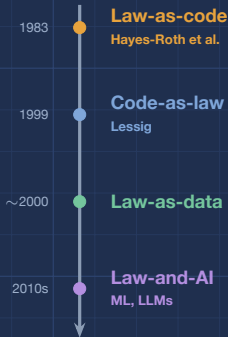
What are computational legal methods (CLM)?

- **Definition:** using computers to model, analyze, and simulate legal phenomena.
- **Motivation:** take on problems that are too complex or too large for other approaches.
- **Related fields:** empirical legal studies, legal informatics, AI & law, legal computer science (statistics, networks, NLP, ML), legal tech.



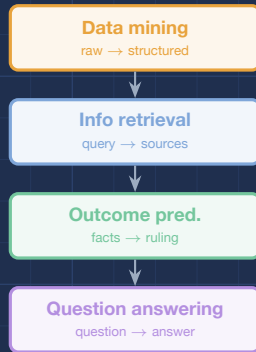
A short history

- **Law-as-code:** legal knowledge in computable form (Hayes-Roth et al. 1983; Sergot & Specen 1986).
- **Code-as-law:** code as a governance tool (Lessig 1999).
- **Law-as-data:** ~2000, probabilistic, data-driven approaches to law.
- **Law-and-AI:** 2010s, ML and LLMs generate knowledge about law.



Main problems

- **Data mining:** build large, high-quality datasets of law.
- **Legal information retrieval:** find relevant legal information.
- **Outcome prediction:** predict case outcomes.
- **Legal question answering:** generate correct statements about the law.



Main approaches

- **Symbolic / logic-based:** formal logic, symbolic AI.
- **Rule-based:** rules and regex over legal text.
- **Graph-theoretic:** exact graph algorithms, community detection.
- **Probabilistic:** topic models, ML, embeddings, probabilistic graph algorithms.

Symbolic

$$p \Rightarrow q$$
$$\forall x L(x)$$

Rule-based

if/then
regex

Graph



Probabilistic



One question, five methods

Which of 2,674 other Swedish Supreme Court judgments is most similar to NJA 2019 s. 684 (“mopedbilen”)?

- Similar cases should be treated the same.
- No two cases are identical; there are always differences.
- Given one case, find relevantly and sufficiently similar cases (a retrieval framing).
- **Ranking problem** (what is the most similar case?) or **threshold problem** (which cases are at least X% similar?).

NO SINGLE MEASUREDIFFERENT MEASURES HAVE DIFFERENT PROS AND CONS

The corpus

2,675

HD CASES1981-2022

215,550

PARAGRAPHS OF FULL

TEXT $\approx$ 86 MB

46,595

STRUCTURED REFERENCES

8,629

HD→HD CITATIONS

(5,688 IN-CORPUS)

- **Composition:** 1,474 civil (T), 1,200 criminal (B), 1 administrative (P).
- **Text structure:** each opinion is pre-coded into sections — intro, background, actor presentation, reasoning, verdict.
- **References:** every citation is classified (statute, case, legislative material, principle, literature, ...).
- **Source:** the sehc case-law database. You receive flat CSV files --- no database access required.

One case, raw

```
CASE    B 1064-19 .....NJA 2019 s. 684
DATE    2019-07-12 ..CLASS B (criminal) ..AREA 8
KW       Rattfylleri, Moped, Mopedbil
```

[intro] Allmän åklagare väckte vid Skaraborgs tingsrätt åtal mot L.B. för grovt rattfylleri...

L.B. har kört mopedbil efter att ha druckit alkoholhaltiga drycker... 0,51 mg/l utandningsluft...

[reasoning] 16. Den mopedbil som L.B. körde "är en moped klass I. Som utgångspunkt ska rattfylleri med ett sådant fordon anses vara av normalgraden...

- The court's own summary header — a human-written “feature”.
- Court-provided keywords and class (B = criminal).
- Full text in Swedish, *pre-structured* into sections.
- 96 paragraphs; this case also contains a dissent (*skiljaktig*).
- This structure is what lets us study opinions computationally.

How the live demos work

DATA

4 CSV files + a prompt

CODE

3 R scripts

OUTPUT

plots & tables

- No R experience required — I drive, you follow the printed script.
- Every result is computed live from the CSVs you receive — no pre-cooked charts.
- One corpus and one question throughout — similarity to the anchor case, measured six ways.

One question, five answers — the design choices

DIMENSIONS

Which similarity? facts, reasoning, cited sources

MEASUREMENTS

How to turn law into numbers? word counts, vectors, graph distances, prompt outputs

ABSTRACTNESS

How coarse? court codes and area labels --- or unstructured court text

EVALUATION

Which answer is right? no gold standard --- compare the lists, audit the reasons

EVERY METHOD IS A CHAIN OF CHOICES --- EACH WITH PROS AND CONS

FIELD GUIDE // END OF SHEET

STOP // QUESTIONS?

NEXTPART 02 // QUANTITATIVE TEXT ANALYSIS
(AFTER THE BREAK)

